ELMFIRE Validation Guide

Authors

Yiren Qin
Nikolaos Kalogeropoulos
Daniel San Martin
Adam Laird
Dwi M. J. Purnomo
Mateo Giorgi
Maria Theodore
Kasra Shamsaei
Majid Bavandpour
Arnaud Trouvé
Michael J. Gollner
Chris Lautenberger

Affiliations

University of California, Berkeley
Cloudfire Inc.
University of Maryland, College Park

ELMFIRE version: 2025.1002
Compiled: October 18, 2026

About This Guide

This guide compares ELMFIRE predictions with observations for the active validation cases. Each report describes the event, input data, modeling choices, comparison methods, and uncertainties. The opening summary presents the calculated decisions and the computational information available during report preparation. A completed simulation does not by itself establish validation.

Contents

About This Guide 1

1 How to Read This Guide 3

2 Validation summary 4
- 2.1 Computational environment recorded during report preparation . . . 4
- 2.2 Comparison type and declared parallel allocation . . . 4
- 2.3 Scientific decision summary . . . 4

3 Common landscape-validation evidence 5

4 camp_fire – 2018 Camp Fire landscape reconstruction 6

5 thomas_fire – 2017 Thomas Fire landscape reconstruction 16

6 tubbs_fire – 2017 Tubbs Fire landscape reconstruction 26

1 How to Read This Guide

Start with the validation summary for the recorded computational environment, declared parallel allocation, and scientific decision table. The case identifiers in the summary link to the corresponding case-report entries. The common landscape-validation evidence then explains the metrics and limitations shared by the current cases, followed by each complete standalone case report with its event-specific provenance, configuration, figures, results, and limitations.

Interpret PASS and FAIL only where a case defines a justified acceptance decision. CHARACTERIZED means that descriptive metrics exist without a justified pass threshold. Missing, unrun, incomplete, or otherwise non-decisional evidence is shown as NOT EVALUABLE. A completed narrative alone does not establish validation evidence. The environment recorded during report preparation is not necessarily the execution environment of every included case.

2 Validation summary

2.1 Computational environment recorded during report preparation

The following information describes the environment available during preparation of this guide. It does not establish the environment used for earlier simulations. An available compiler is not necessarily the compiler used to build the model. ELMFIRE was not executed to obtain these observations; an undeclared model release remains unknown.

Configuration	Recorded value
Recorded at (Coordinated Universal Time)	2026-10-18T17:19:41+00:00
Declared ELMFIRE release	2025.1002
Operating system	Linux-7.0.12-linuxkit-aarch64-with-glibc2.35
Processor architecture	aarch64
Message Passing Interface (MPI) implementation	mpirun (Open MPI) 4.1.2
Available Fortran compiler	GNU Fortran (Ubuntu 11.4.0-1ubuntu1~22.04.3) 11.4.0
Python version	CPython 3.10.12
Scientific analysis libraries	NumPy 1.26.4; Matplotlib 3.8.4; PyYAML 6.0.3; Rasterio 1.4.4; GDAL 3.4.1; Pandas 2.2.2; GeoPandas 1.1.4; Shapely 2.1.2

2.2 Comparison type and declared parallel allocation

The comparison type distinguishes isolated and coupled processes or the spatial scale of validation. The process count is the allocation declared for each case, not a measurement of resources used in an earlier simulation. Missing declarations are not inferred.

Case	Comparison type	MPI processes
camp_fire	Landscape-scale validation	50
thomas_fire	Landscape-scale validation	50
tubbs_fire	Landscape-scale validation	50

2.3 Scientific decision summary

The decisions below follow the established calculated metrics and case-specific acceptance criteria. Completing a simulation or preparing a report does not establish PASS. Missing or insufficient scientific evidence is reported as NOT EVALUABLE.

Total cases: 3 **PASS:** 0 **FAIL:** 0 **CHARACTERIZED:** 3 **NOT EVALUABLE:** 0

Case	Category	Title	Decision
camp_fire	Landscape-scale validation	2018 Camp Fire landscape reconstruction	**CHARACTERIZED**
thomas_fire	Landscape-scale validation	2017 Thomas Fire landscape reconstruction	**CHARACTERIZED**
tubbs_fire	Landscape-scale validation	2017 Tubbs Fire landscape reconstruction	**CHARACTERIZED**

Decision vocabulary. PASS and FAIL are case-defined scientific decisions. CHARACTERIZED denotes descriptive validation metrics without a justified acceptance threshold. NOT EVALUABLE denotes missing, unrun, incomplete, unsupported, or otherwise non-decisional evidence.

3 Common landscape-validation evidence

The current Camp, Thomas, and Tubbs landscape cases were reconstructed from the supplied historical-fire datasets. Each is designed to compare a 50-member ELMFIRE ensemble with cumulative Visible Infrared Imaging Radiometer Suite (VIIRS) active-fire detections. The case reports retain the event-specific input provenance, namelist rationale, evidence status, and unresolved limitations.

Their common spatial metrics are Jaccard overlap, precision, recall, F_1, predicted-to-observed area ratio, and Brier score. Together these distinguish spatial mismatch, overgrowth, undergrowth, scale bias, and probabilistic error. Because the recovered archives define no acceptance thresholds and do not quantify VIIRS-hull uncertainty, completed runs are reported as CHARACTERIZED rather than PASS or FAIL unless justified thresholds are declared before inspecting results. Unresolved model-input provenance, VIIRS collection or granule provenance, and event-specific calibration history are stated explicitly in the standalone reports.

2018 Camp Fire Landscape Reconstruction

1 Validation objective and scope

This case asks whether ELMFIRE can reconstruct the landscape-scale spatial growth of the 2018 Camp Fire Landscape Reconstruction through 36 hours when driven by the supplied landscape, weather, moisture, ignition, spotting, and wildland–urban interface (WUI) inputs. The comparison evidence is a time-filtered set of active-fire detections from the Visible Infrared Imaging Radiometer Suite (VIIRS). The case exercises the coupled surface- and crown-fire, level-set, firebrand, ember-ignition, Monte Carlo, and building-spread processes. It does not isolate or verify any single algorithm, and it does not establish predictive skill for events outside the three received reconstructions.

The scientific question is deliberately two-sided: does the ensemble overlap the observation-derived footprint, and does it do so without achieving overlap by excessive fire growth? Overlap, omission, commission, area, and probabilistic error are therefore reported together. The archive provides no independent, predeclared acceptance thresholds, so completed runs are *characterized*; they are not labeled PASS or FAIL.

2 Reference event, experiment, and observations

The configured event start, expressed in Pacific standard time (PST), is 8 November 2018 at 14:00 PST. The comparison instant is 10 November 2018 at 02:00 PST (10:00 UTC), with UTC indicating Coordinated Universal Time. The supplied observation dataset contains 2260 point records in the geographic coordinate system defined by EPSG:4326; 2017 occur on or before the comparison cutoff. The observation attributes identify the instrument as VIIRS and are consistent with the National Aeronautics and Space Administration (NASA) Fire Information for Resource Management System (FIRMS) active-fire product. NASA describes VIIRS active-fire data as nominal 375-m thermal detections, not mapped fire perimeters. General product information is available from NASA FIRMS active-fire product documentation and NASA Earthdata VIIRS near-real-time product documentation.

The exact FIRMS collection, granules, download URL, acquisition date, filtering history, and license are not recorded in the case material. Consequently the NASA attribution is inferred from the observation attributes rather than established by complete provenance. Integrity checks identify the supplied observations, which remain unchanged during preprocessing.

For raster comparison, detections through the cutoff are projected to the simulation coordinate reference system (CRS) and enclosed by a concave hull with ratio 0.2 and no holes. If the installed geometry library lacks that operation, the postprocessor reports and uses a convex-hull fallback. This converts intermittent points into a comparison support; it is a declared comparison assumption, not an observed final perimeter. Cloud, smoke, overpass timing, geolocation error, false detections, and the hull choice remain observation uncertainties.

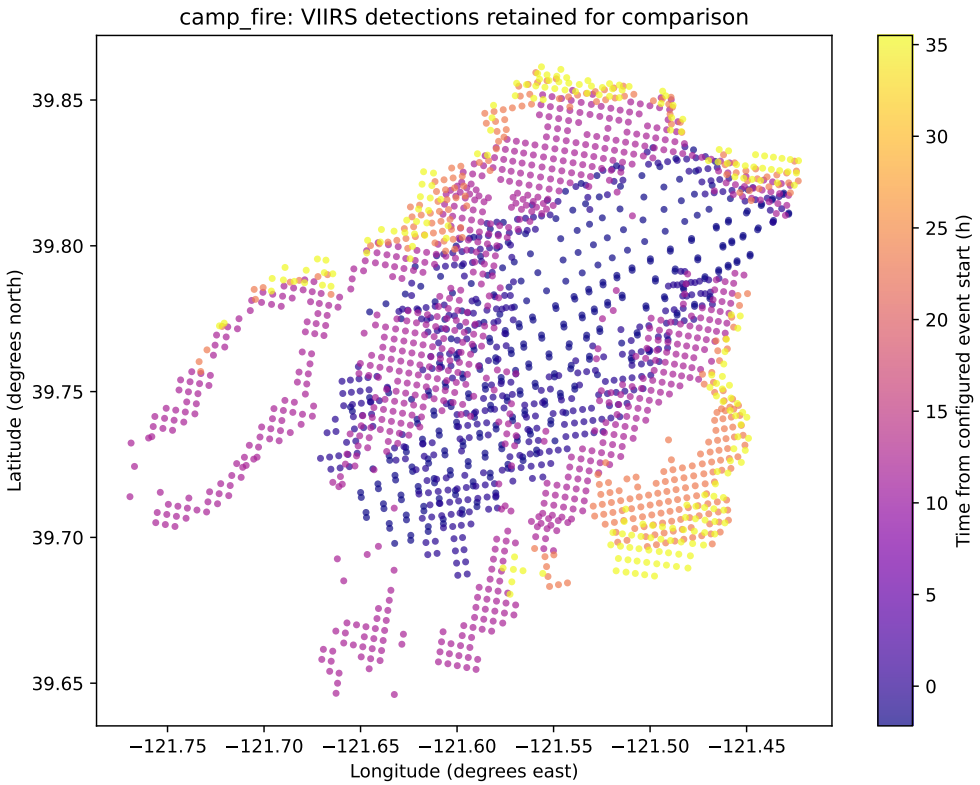


Figure 1: Supplied VIIRS detections retained through the comparison cutoff, colored by elapsed time from the configured event start.

3 Input data provenance and characterization

The supplied historical-fire reconstruction is identified by the SHA-256 checksum `0b0e8f7ce3bbea4665269ff7f5b88dd373f3d9210b3fa4ddc7916b610f724033`. The extracted source data are retained unchanged, and their identities are checked before analysis. The collection's only supplied provenance note is "From Dwi." Upstream product names, providers, vintages, acquisition dates, processing history, and licenses for fuels, terrain, meteorology, moisture, WUI, and ignition data are unknown. The report therefore describes the archived values but does not infer unsupported product identities such as LANDFIRE or a specific weather analysis.

Field	Minimum	Mean	Standard deviation	Maximum
Elevation	30	470.9	365.7	1620
Slope	0	11.74	10.32	63
Wind speed	0	8.313	6.306	33.35
1-h moisture	6	6	0	6
10-h moisture	7	7	0	7
100-h moisture	8	8	0	8

Table 1: Input statistics over valid cells and the 25 weather bands selected by the namelist. Wind speed is interpreted as 20-ft mph and dead-fuel moisture as percent from the archived flags and inspected current source; upstream metadata that would independently confirm these units is absent.

No additional active-range anomaly was identified by the structural and statistical checks; upstream plausibility remains unverified because product metadata are absent.

Aspect and wind direction are circular quantities. Their arithmetic means are not used as directional summaries; the circular means are 329.6° and 2.253°, respectively. Exact means and standard deviations use every valid selected value. Reported quantiles use a deterministic nearest-neighbor sample of at most 512×512 cells per band, and are labeled accordingly.

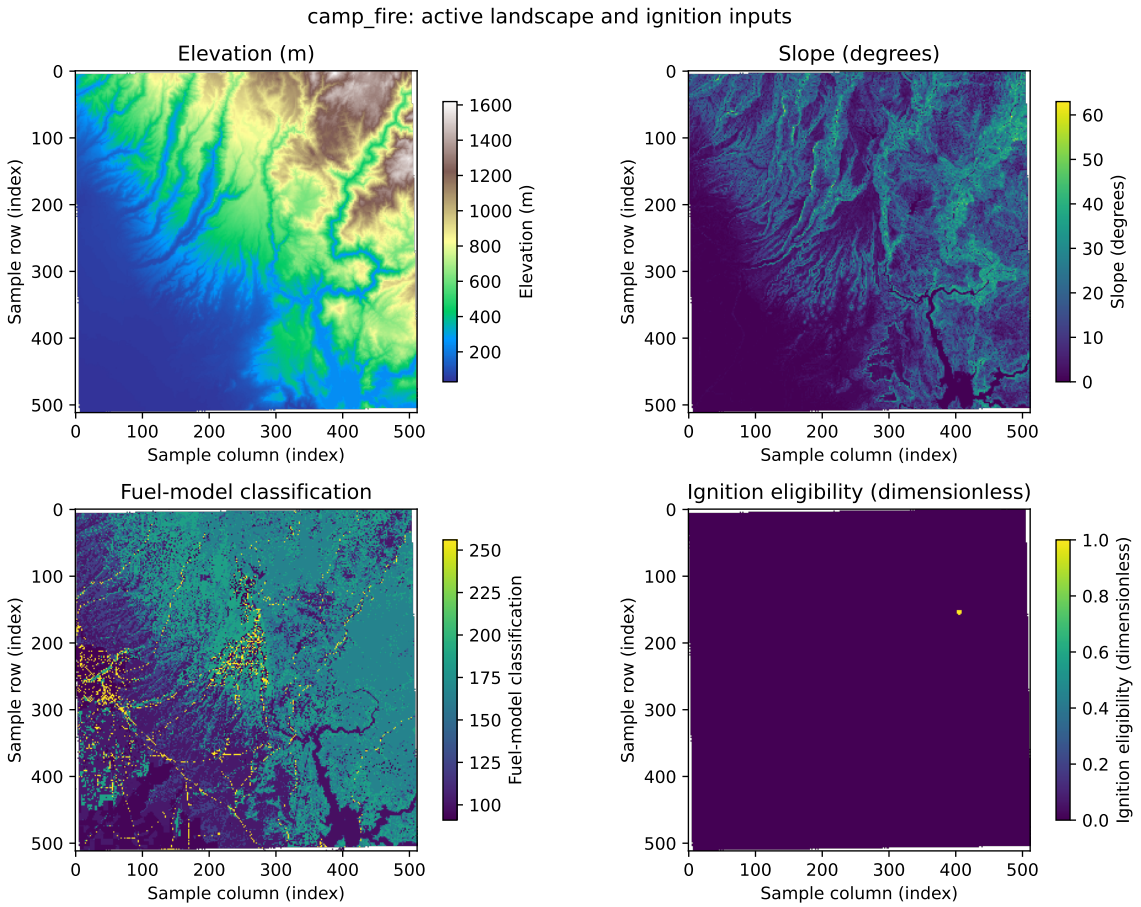


Figure 2: Active elevation, slope, fuel-model, and ignition-mask inputs. These plots characterize model forcing; they are not validation evidence.

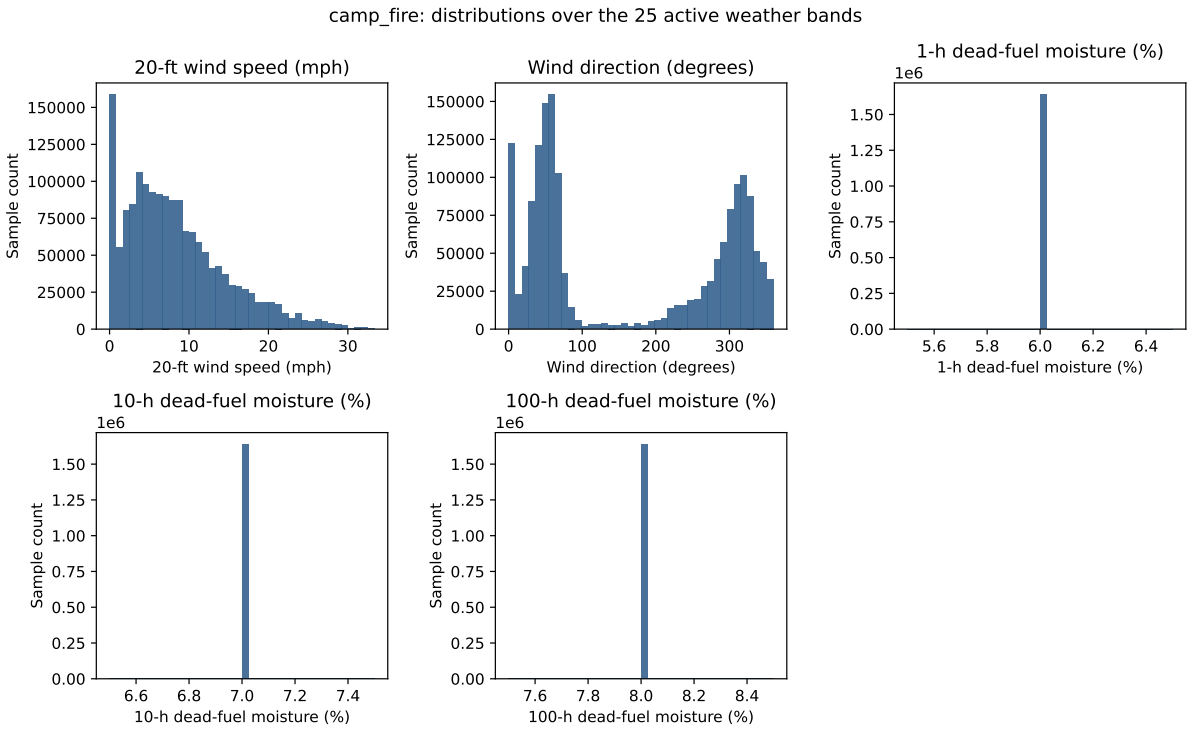


Figure 3: Distributions sampled deterministically from the 25 weather and dead-fuel-moisture bands selected by the namelist.

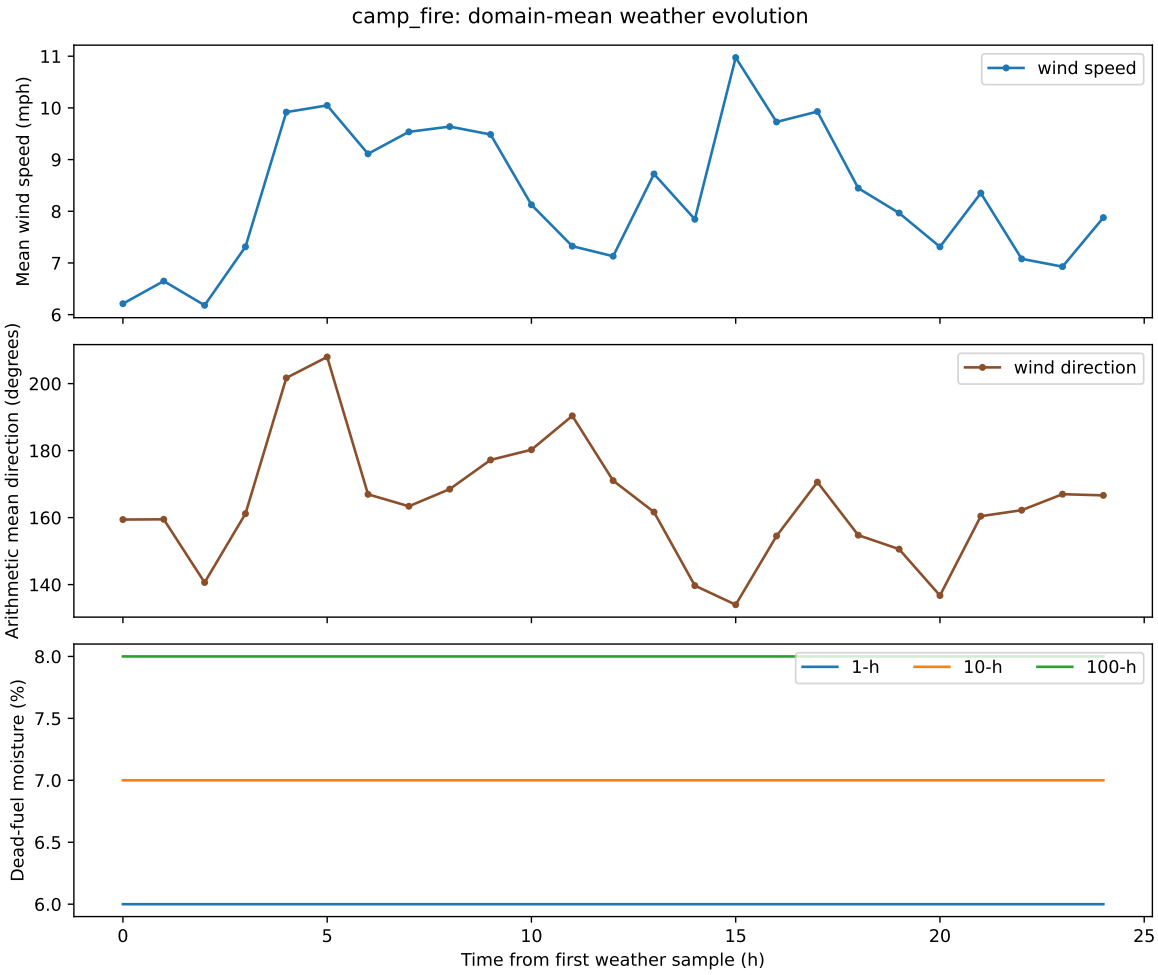


Figure 4: Exact domain-mean wind and dead-fuel-moisture values by active band.

4 Model configuration and rationale

The active domain is 1457 by 1476 cells at 30 m in EPSG:26910. Weather is 37-band weather rasters; bands 1–25 are selected at one-hour intervals, and the ignition treatment is random ignition sampling from the archived ignition mask. The simulation uses 50 members, a target Courant–Friedrichs–Lewy (CFL) number of 0.45, crown-fire model 2, and fire-power-proportional firebrand generation with empirical transport distance, Eulerian accumulation, and physical ignition. These physical values, seeds, uncertainty ranges, ignition choices, and WUI settings are retained from the received archive. In the available material they are *documented by configuration* but their scientific selection rationale is unsupported; they must not be described as independently derived or literature-justified.

The original settings were adapted only where required for compatibility with the inspected ELMFIRE formulation at source revision `efa0b8fa3935ea4cd30a2bd054c7e9b83db81163`, which already contained additional modifications when inspected. Obsolete domain-description and diagnostic settings were omitted. The earlier spotting choices were expressed as fire-power-proportional generation, empirical transport distance, Eulerian accumulation, and physical ignition. The required physical values are retained in the case specification; this compatibility review does not establish their scientific justification.

Choice	Evidence class	Basis and limitation
Landscape, weather, ignition, Monte Carlo, spotting, and WUI values	documented	Retained from the received archive; upstream derivation and event-specific tuning history are unknown.
Model-setting compatibility	derived	Required by the choices supported by the inspected model formulation; no physical values were tuned.
Comparison time	documented	Fixed in the case specification and consistent with the supplied reconstruction procedure; original scientific rationale is absent.
$p \geq 0.50$ footprint	selected for this comparison	Majority-ensemble interpretation used for reproducibility; it is not an acceptance threshold and sensitivity should be reported for scientific use.
VIIRS hull ratio 0.2	selected for this comparison	Compact cumulative support for point detections; not calibrated to an uncertainty-bounded perimeter.

Table 2: Provenance of consequential configuration and comparison choices.

5 Preprocessing and reproducibility

Preprocessing verifies the supplied reconstruction and observations by size and SHA-256 checksum. It checks that required landscape rasters share a grid, confirms that the namelist-selected weather rasters contain at least 25 bands, computes input and observation statistics, and regenerates the four input figures. The resulting statistics and figures characterize the source data without modifying the original evidence.

Simulation is followed by analysis of the complete ensemble. The declared default allocation is 50 Message Passing Interface (MPI) processes, one per ensemble member, without an additional coordinating process. The original simulation results remain available for examination. Reproducible input identity does not by itself guarantee result reproducibility: each run must also record the actual ELMFIRE revision, compiler, dependencies, and platform.

6 Validation metrics and comparison method

For member i, let $\tau_i(\mathbf{x})$ be the time-of-arrival raster and t_c the declared comparison time. The ensemble burn probability and thresholded footprint are

$$p(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[0 < \tau_i(\mathbf{x}) \leq t_c] \,, \qquad P = \{\mathbf{x} : p(\mathbf{x}) \geq 0.50\}. \tag{1}$$

For the rasterized observation support O, intersection $I = |P \cap O|$, and valid-cell count M, the analysis reports

$$J = \frac{|P \cap O|}{|P \cup O|}, \qquad \text{precision} = \frac{I}{|P|}, \qquad \text{recall} = \frac{I}{|O|}, \tag{2}$$

$$F_1 = \frac{2\,\text{precision}\,\text{recall}}{\text{precision} + \text{recall}}, \qquad R_A = \frac{|P|}{|O|}, \qquad \text{Brier} = \frac{1}{M}\sum_{\mathbf{x}}(p - y)^2. \tag{3}$$

Jaccard is a symmetric footprint-overlap summary. Precision diagnoses commission/overgrowth, recall diagnoses omission/undergrowth, and F_1 balances those errors. The area ratio exposes size compensation that overlap alone can hide. The Brier score retains ensemble probabilities, although the large unburned domain can dominate it. Burned-area growth and conditional median arrival time add temporal context. VIIRS detections and their derived hull are the observational baseline; no previous simulation result is used as truth.

Metrics are calculated only when exactly 50 uniquely identified members reach the comparison time and share one grid and CRS. This completeness requirement prevents partial runs from being reported as scientific results. The static aspect raster defines the analysis-domain mask. Member-specific NoData cells in time-of-arrival outputs represent cells not burned by that member; they are therefore retained as zero burn outcomes rather than treated as a grid-mask mismatch.

7 Simulation results and field-level comparison

The current scientific status is **CHARACTERIZED**. Metrics are descriptive because no independent predeclared acceptance thresholds are available. Metrics and figures are generated only from the complete, current ensemble. If that ensemble is unavailable, the analysis reports the result as not evaluable rather than substituting legacy outputs or zero-valued metrics.

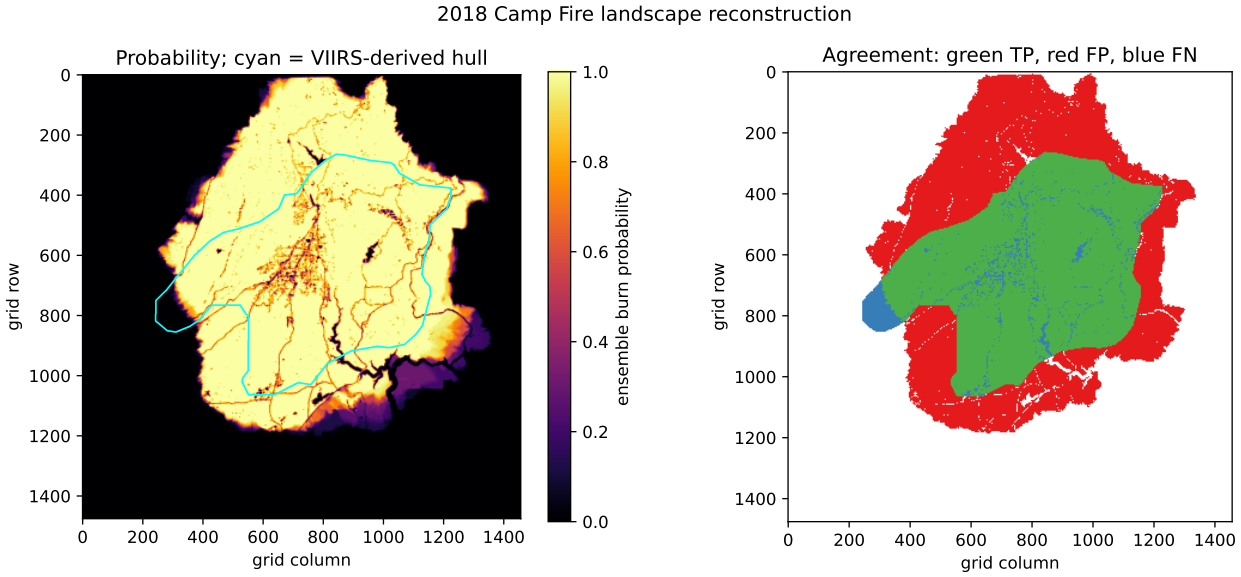


Figure 5: Ensemble burn probability, VIIRS-derived support, and cell-level true-positive, false-positive, and false-negative regions.

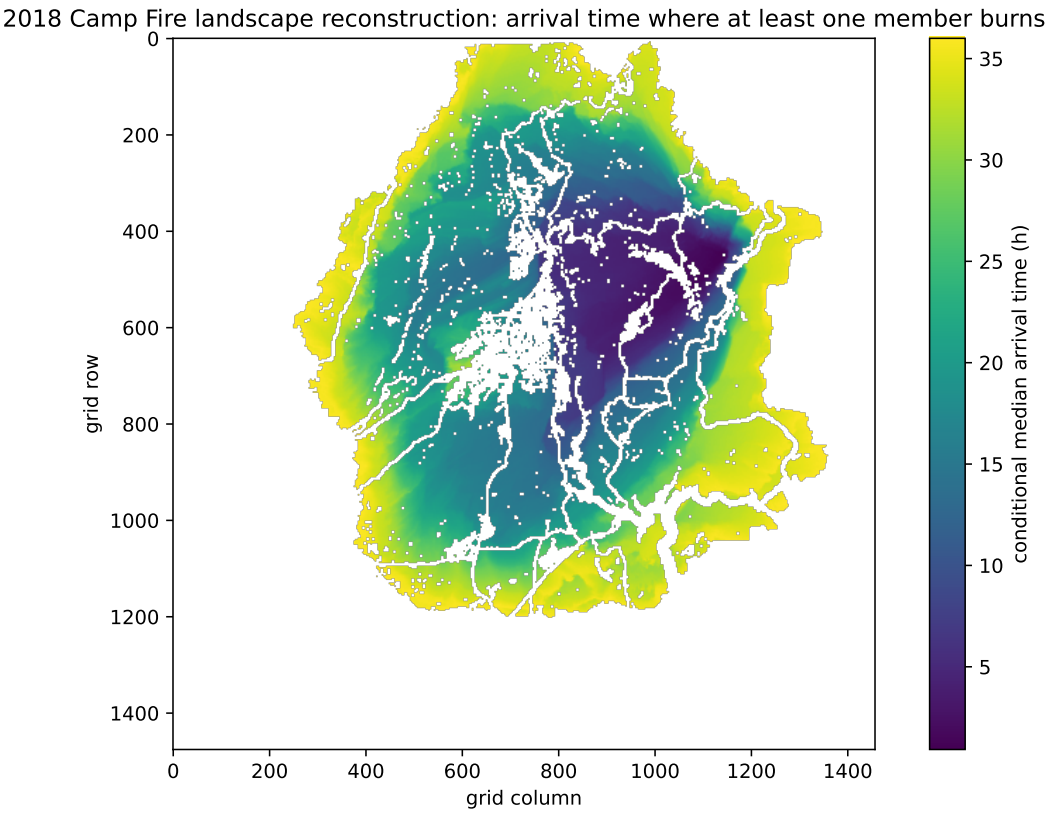


Figure 6: Conditional median arrival time among members that burn each cell by the comparison time. Conditioning is stated because it omits no-burn members.

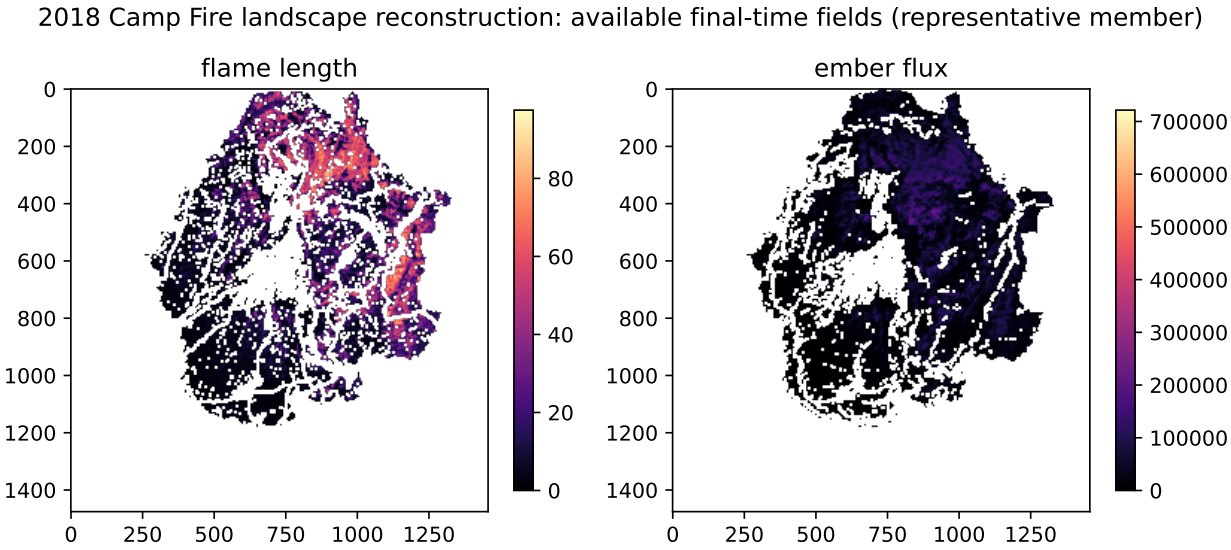


Figure 7: Applicable final-time two-dimensional fields found by the postprocessor. Omission of this panel means no recognized optional field was available; it does not imply a zero field.

The simulation requests flame length and ember flux in addition to arrival time. It does not request a structure-damage count or damage-time output, so a damaged-structures time series is not reported

and must not be inferred from WUI input rasters alone.

8 Integrated statistics and calculated metrics

Metric	Current value
Jaccard overlap	0.4858
Precision	0.5061
Recall	0.9236
F_1 score	0.6539
Brier score	0.197
Predicted/observed area ratio	1.825
Predicted area (km^2)	720.5
Observation-support area (km^2)	394.8
Ensemble members found	50

Table 3: Calculated statistics. **MISSING** is intentionally conspicuous when the completeness requirement prevents evaluation.

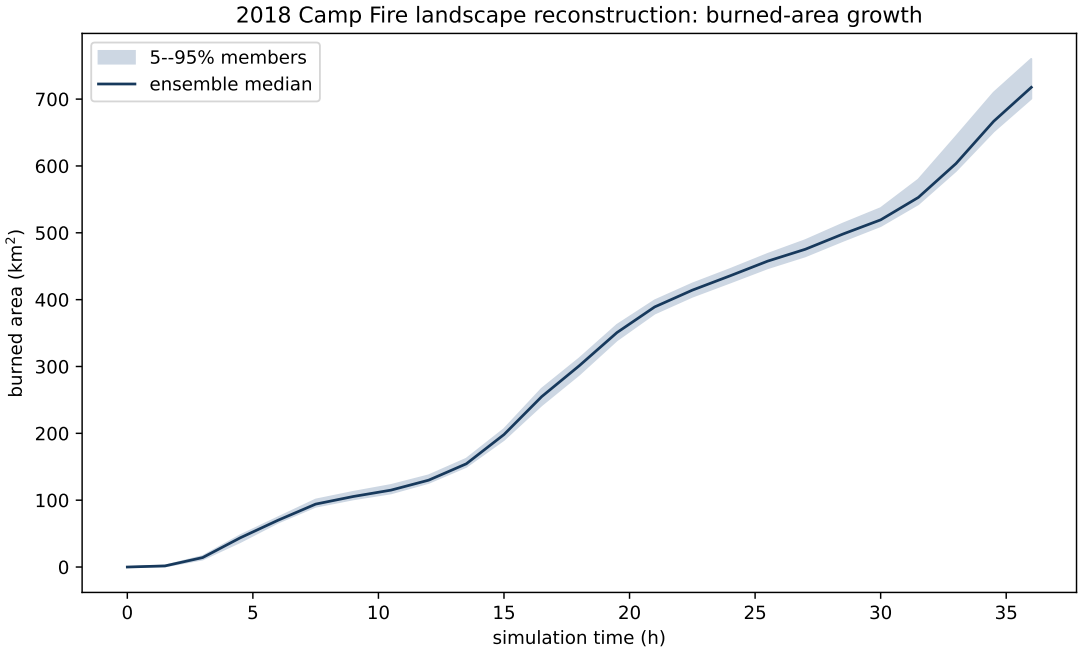


Figure 8: Median and 5th–95th percentile ensemble burned-area growth.

These values are compared with the observation-derived support and not with a stored ELMFIRE baseline. Because neither metric thresholds nor observation uncertainty bounds were declared independently, a populated table supports scientific characterization only.

9 Interpretation and uncertainty

A low Jaccard value with high recall and low precision indicates spatial overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory. An area ratio near one cannot establish spatial agreement. The Brier score must be read with the footprint metrics because correct prediction of the large unburned domain may obscure errors near the fire.

The dominant limitations are incomplete input provenance, an observation product that samples thermal anomalies rather than perimeter, uncertain hull sensitivity, unknown event-specific calibration history, and absence of predeclared acceptance bounds. Model-form error, uncertain fuels and moisture, meteorological representation, ignition timing/location, spotting parameters, and WUI parameters are not separable using these aggregate metrics alone. The 50 archived perturbations characterize one configured ensemble; they do not by themselves quantify all epistemic uncertainty.

10 Conclusions and limitations

This case characterizes the agreement between a complete simulated ensemble and a spatial support derived from active-fire detections. Input statistics describe the forcing, spatial comparisons distinguish omission and commission, and integrated quantities provide context for the overlap metrics. Its current scientific result is **CHARACTERIZED**; the accompanying reason states whether the complete current ensemble supported quantitative characterization.

To support a defensible validation decision, future case owners must supply the upstream input citations and licenses, exact VIIRS collection/granule and acquisition provenance, the derivation or literature basis for ignition, spotting, WUI, and perturbation settings, disclosure of any calibration against this event, and acceptance thresholds fixed before inspecting the evaluated run. Until then, even a completed run is correctly labeled CHARACTERIZED, not PASS or FAIL.

2017 Thomas Fire Landscape Reconstruction

1 Validation objective and scope

This case asks whether ELMFIRE can reconstruct the landscape-scale spatial growth of the 2017 Thomas Fire Landscape Reconstruction through 17 hours when driven by the supplied landscape, weather, moisture, ignition, spotting, and wildland–urban interface (WUI) inputs. The comparison evidence is a time-filtered set of active-fire detections from the Visible Infrared Imaging Radiometer Suite (VIIRS). The case exercises the coupled surface- and crown-fire, level-set, firebrand, ember-ignition, Monte Carlo, and building-spread processes. It does not isolate or verify any single algorithm, and it does not establish predictive skill for events outside the three received reconstructions.

The scientific question is deliberately two-sided: does the ensemble overlap the observation-derived footprint, and does it do so without achieving overlap by excessive fire growth? Overlap, omission, commission, area, and probabilistic error are therefore reported together. The archive provides no independent, predeclared acceptance thresholds, so completed runs are *characterized*; they are not labeled PASS or FAIL.

2 Reference event, experiment, and observations

The configured event start, expressed in Pacific standard time (PST), is 4 December 2017 at 18:30 PST. The comparison instant is 5 December 2017 at 11:30 PST (19:30 UTC), with UTC indicating Coordinated Universal Time. The supplied observation dataset contains 6804 point records in the geographic coordinate system defined by EPSG:4326; 1446 occur on or before the comparison cutoff. The observation attributes identify the instrument as VIIRS and are consistent with the National Aeronautics and Space Administration (NASA) Fire Information for Resource Management System (FIRMS) active-fire product. NASA describes VIIRS active-fire data as nominal 375-m thermal detections, not mapped fire perimeters. General product information is available from NASA FIRMS active-fire product documentation and NASA Earthdata VIIRS near-real-time product documentation.

The exact FIRMS collection, granules, download URL, acquisition date, filtering history, and license are not recorded in the case material. Consequently the NASA attribution is inferred from the observation attributes rather than established by complete provenance. Integrity checks identify the supplied observations, which remain unchanged during preprocessing.

For raster comparison, detections through the cutoff are projected to the simulation coordinate reference system (CRS) and enclosed by a concave hull with ratio 0.2 and no holes. If the installed geometry library lacks that operation, the postprocessor reports and uses a convex-hull fallback. This converts intermittent points into a comparison support; it is a declared comparison assumption, not an observed final perimeter. Cloud, smoke, overpass timing, geolocation error, false detections, and the hull choice remain observation uncertainties.

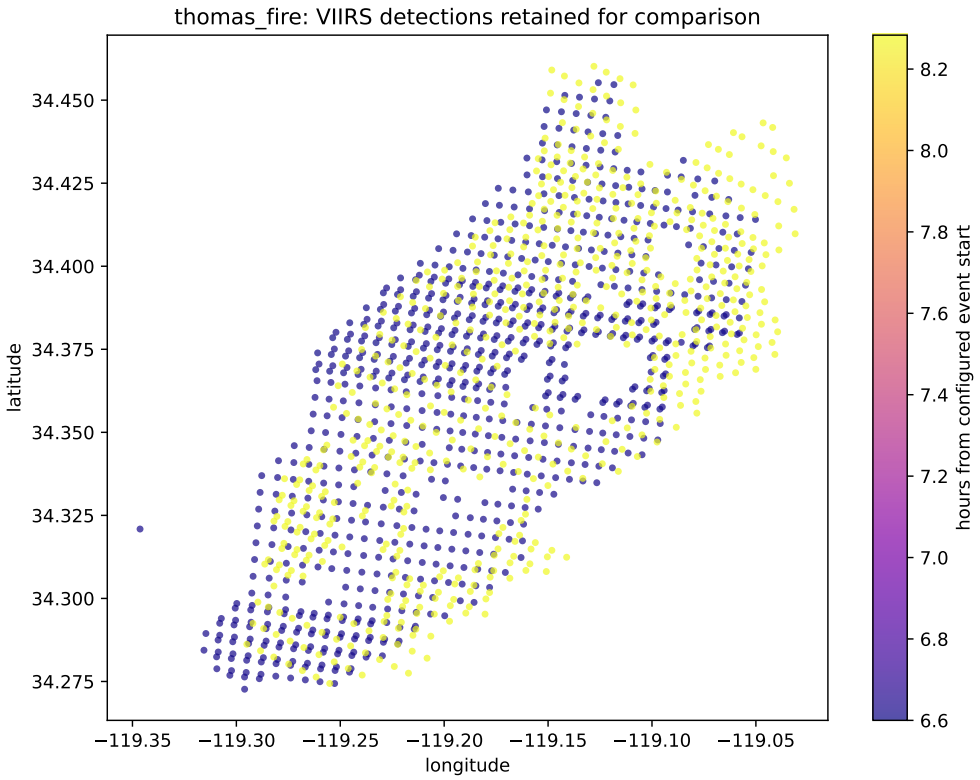


Figure 1: Supplied VIIRS detections retained through the comparison cutoff, colored by elapsed time from the configured event start.

3 Input data provenance and characterization

The supplied historical-fire reconstruction is identified by the SHA-256 checksum `642689c45406337886a8dea52e08ff413fd3e7c2a933dcc6e0f7f07954a3a044`. The extracted source data are retained unchanged, and their identities are checked before analysis. The collection's only supplied provenance note is "From Dwi." Upstream product names, providers, vintages, acquisition dates, processing history, and licenses for fuels, terrain, meteorology, moisture, WUI, and ignition data are unknown. The report therefore describes the archived values but does not infer unsupported product identities such as LANDFIRE or a specific weather analysis.

Field	Minimum	Mean	Standard deviation	Maximum
Elevation	-1e+04	-1007	3832	2286
Slope	-1e+04	-1502	3591	68
Wind speed	0	26.46	12.74	62.97
1-h moisture	1.301	5.103	2.91	38.25
10-h moisture	2.696	6.503	2.641	15.76
100-h moisture	5.704	10.52	4.272	20.31

Table 1: Input statistics over valid cells and the 25 weather bands selected by the namelist. Wind speed is interpreted as 20-ft mph and dead-fuel moisture as percent from the archived flags and inspected current source; upstream metadata that would independently confirm these units is absent.

The active elevation raster contains values of -10000 even though its declared nodata value is -9999. The statistics retain those cells exactly as ELMFIRE sees them; the source owner must determine whether -10000 is a valid value or an undocumented sentinel before scientific interpretation.

Aspect and wind direction are circular quantities. Their arithmetic means are not used as directional summaries; the circular means are 44.71° and 48.16°, respectively. Exact means and standard deviations use every valid selected value. Reported quantiles use a deterministic nearest-neighbor sample of at most 512×512 cells per band, and are labeled accordingly.

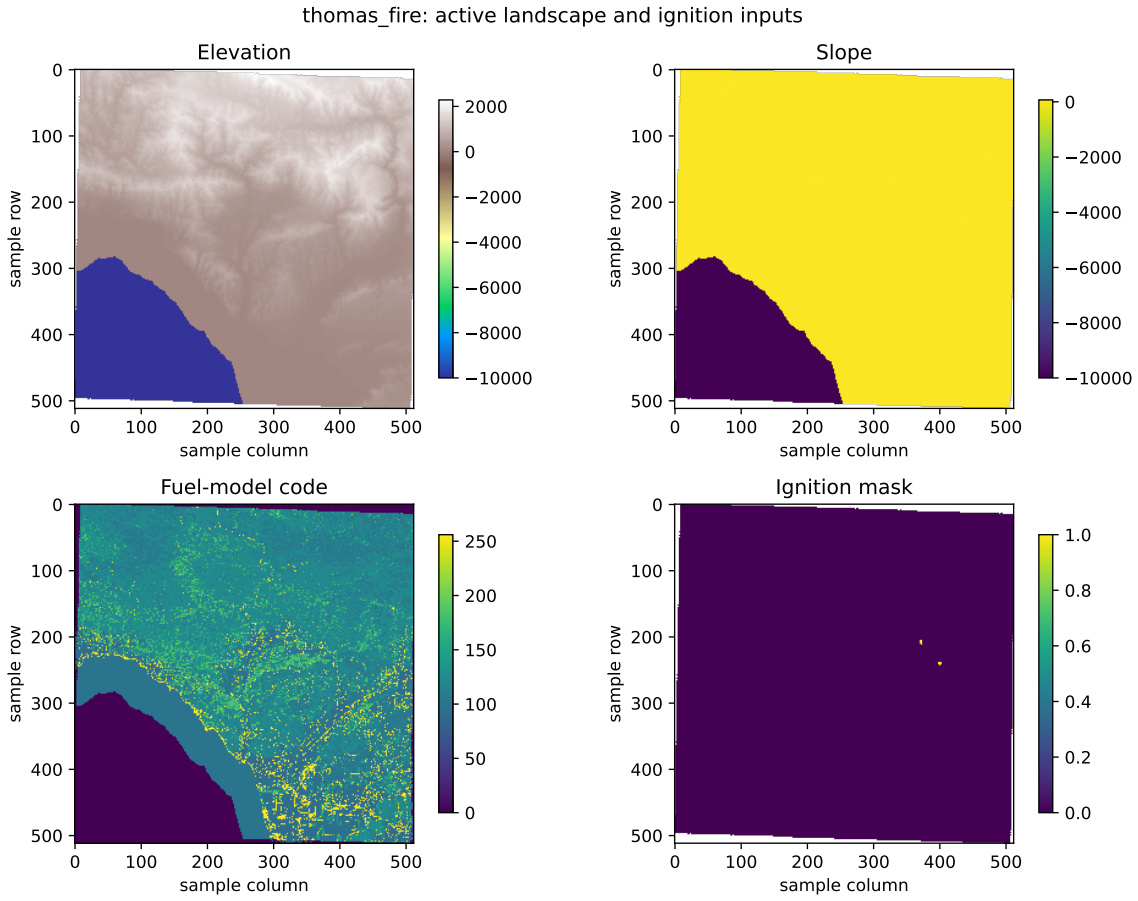


Figure 2: Active elevation, slope, fuel-model, and ignition-mask inputs. These plots characterize model forcing; they are not validation evidence.

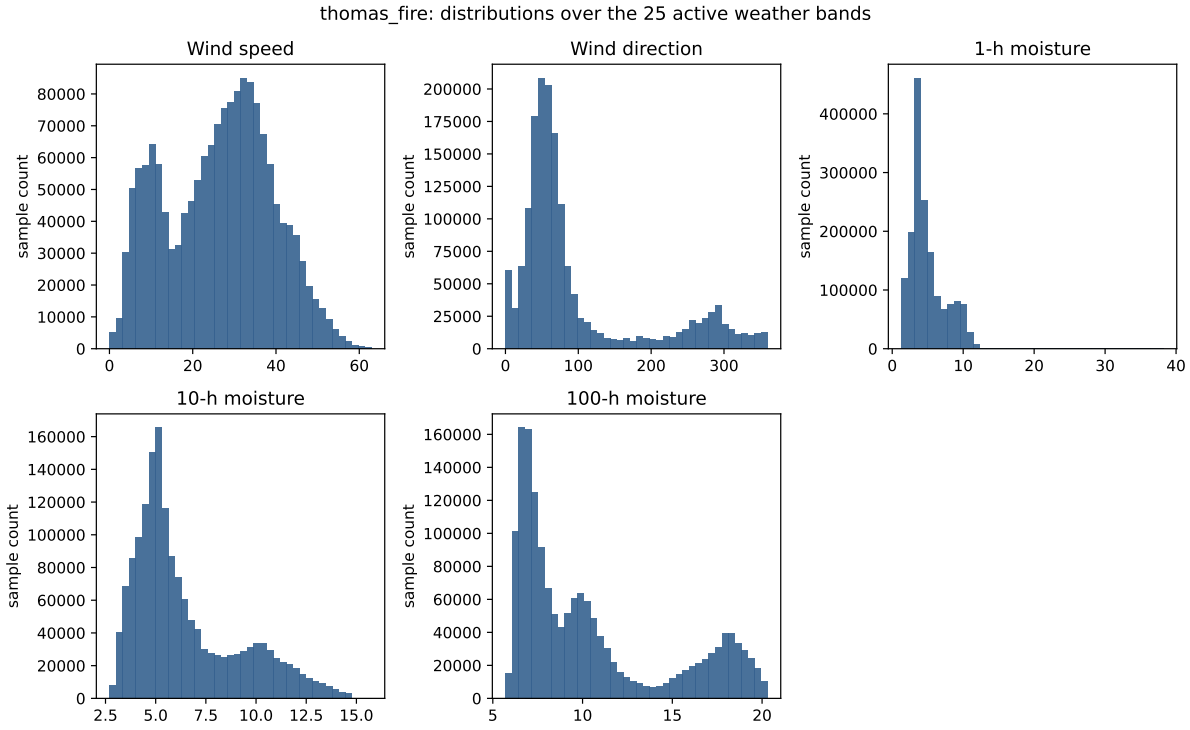


Figure 3: Distributions sampled deterministically from the 25 weather and dead-fuel-moisture bands selected by the namelist.

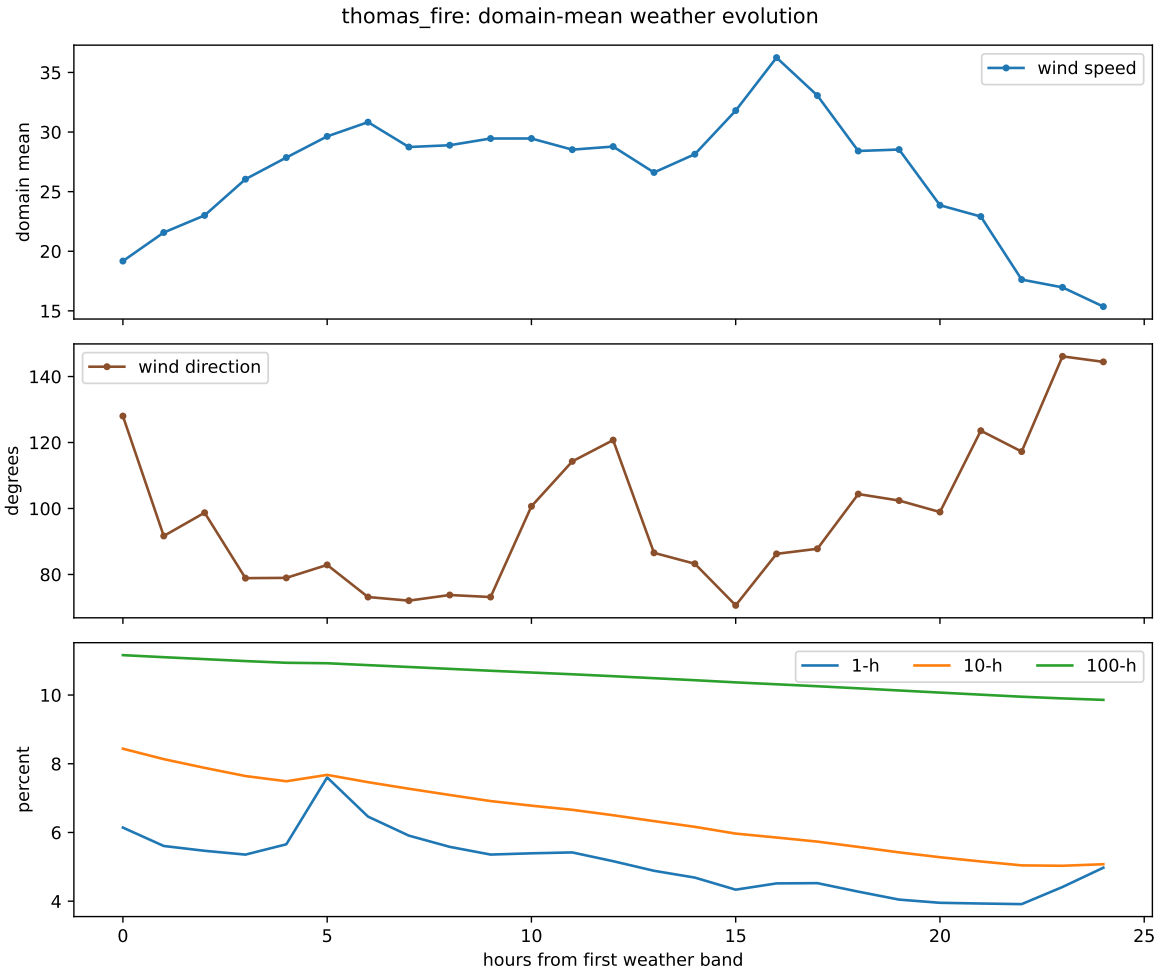


Figure 4: Exact domain-mean wind and dead-fuel-moisture values by active band.

4 Model configuration and rationale

The active domain is 2491 by 1779 cells at 30 m in EPSG:26911. Weather is 29-band weather rasters; bands 1–25 are selected at one-hour intervals, and the ignition treatment is two prescribed point ignitions at archived coordinates and times. The simulation uses 50 members, a target Courant–Friedrichs–Lewy (CFL) number of 0.45, crown-fire model 2, and fire-power-proportional firebrand generation with empirical transport distance, Eulerian accumulation, and physical ignition. These physical values, seeds, uncertainty ranges, ignition choices, and WUI settings are retained from the received archive. In the available material they are *documented by configuration* but their scientific selection rationale is unsupported; they must not be described as independently derived or literature-justified.

The original settings were adapted only where required for compatibility with the inspected ELMFIRE formulation at source revision `efa0b8fa3935ea4cd30a2bd054c7e9b83db81163`, which already contained additional modifications when inspected. Obsolete domain-description and diagnostic settings were omitted. The earlier spotting choices were expressed as fire-power-proportional generation, empirical transport distance, Eulerian accumulation, and physical ignition. Archived building spread type 3 was mapped to current University of California, Berkeley/University of Maryland building-spread formulation (type 2). The required physical values are retained in the case specification; this compatibility review does not establish their scientific justification.

Choice	Evidence class	Basis and limitation
Landscape, weather, ignition, Monte Carlo, spotting, and WUI values	documented	Retained from the received archive; upstream derivation and event-specific tuning history are unknown.
Model-setting compatibility	derived	Required by the choices supported by the inspected model formulation; no physical values were tuned.
Comparison time	documented	Fixed in the case specification and consistent with the supplied reconstruction procedure; original scientific rationale is absent.
$p \geq 0.50$ footprint	selected for this comparison	Majority-ensemble interpretation used for reproducibility; it is not an acceptance threshold and sensitivity should be reported for scientific use.
VIIRS hull ratio 0.2	selected for this comparison	Compact cumulative support for point detections; not calibrated to an uncertainty-bounded perimeter.

Table 2: Provenance of consequential configuration and comparison choices.

5 Preprocessing and reproducibility

Preprocessing verifies the supplied reconstruction and observations by size and SHA-256 checksum. It checks that required landscape rasters share a grid, confirms that the namelist-selected weather rasters contain at least 25 bands, computes input and observation statistics, and regenerates the four input figures. The resulting statistics and figures characterize the source data without modifying the original evidence.

Simulation is followed by analysis of the complete ensemble. The declared default allocation is 50 Message Passing Interface (MPI) processes, one per ensemble member, without an additional coordinating process. The original simulation results remain available for examination. Reproducible input identity does not by itself guarantee result reproducibility: each run must also record the actual ELMFIRE revision, compiler, dependencies, and platform.

6 Validation metrics and comparison method

For member i, let $\tau_i(\mathbf{x})$ be the time-of-arrival raster and t_c the declared comparison time. The ensemble burn probability and thresholded footprint are

$$p(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[0 < \tau_i(\mathbf{x}) \leq t_c] \,, \qquad P = \{\mathbf{x} : p(\mathbf{x}) \geq 0.50\}. \tag{1}$$

For the rasterized observation support O, intersection $I = |P \cap O|$, and valid-cell count M, the analysis reports

$$J = \frac{|P \cap O|}{|P \cup O|}, \qquad \text{precision} = \frac{I}{|P|}, \qquad \text{recall} = \frac{I}{|O|}, \tag{2}$$

$$F_1 = \frac{2\,\text{precision}\,\text{recall}}{\text{precision} + \text{recall}}, \qquad R_A = \frac{|P|}{|O|}, \qquad \text{Brier} = \frac{1}{M}\sum_{\mathbf{x}}(p - y)^2. \tag{3}$$

Jaccard is a symmetric footprint-overlap summary. Precision diagnoses commission/overgrowth, recall diagnoses omission/undergrowth, and F_1 balances those errors. The area ratio exposes size compensation that overlap alone can hide. The Brier score retains ensemble probabilities, although the large unburned domain can dominate it. Burned-area growth and conditional median arrival time add temporal context. VIIRS detections and their derived hull are the observational baseline; no previous simulation result is used as truth.

Metrics are calculated only when exactly 50 uniquely identified members reach the comparison time and share one grid and CRS. This completeness requirement prevents partial runs from being reported as scientific results. The static aspect raster defines the analysis-domain mask. Member-specific NoData cells in time-of-arrival outputs represent cells not burned by that member; they are therefore retained as zero burn outcomes rather than treated as a grid-mask mismatch.

7 Simulation results and field-level comparison

The current scientific status is **CHARACTERIZED**. Metrics are descriptive because no independent predeclared acceptance thresholds are available. Metrics and figures are generated only from the complete, current ensemble. If that ensemble is unavailable, the analysis reports the result as not evaluable rather than substituting legacy outputs or zero-valued metrics.

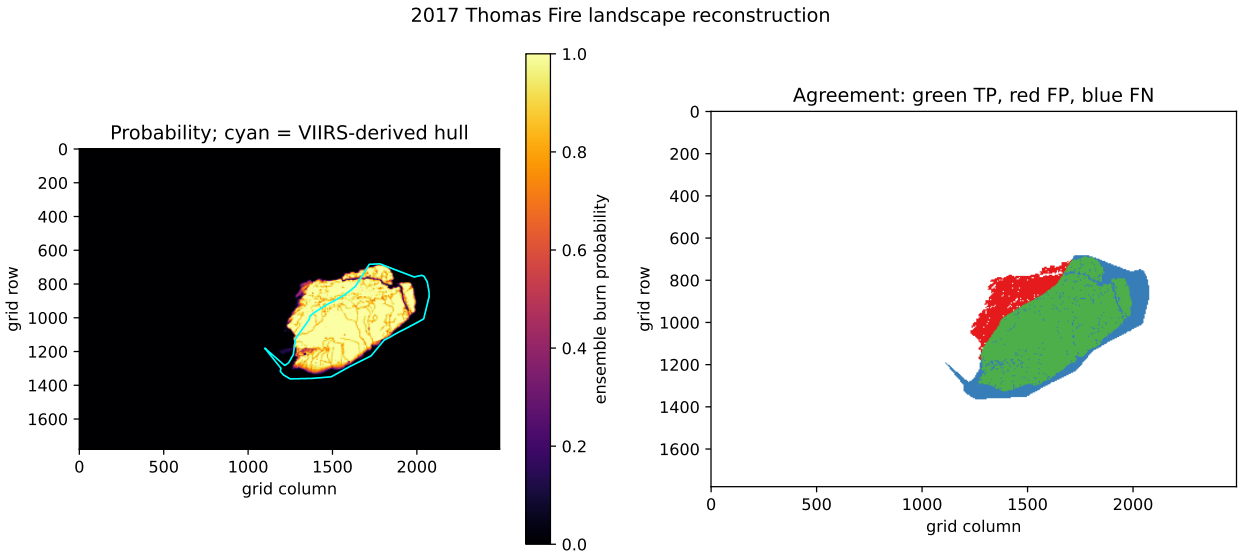


Figure 5: Ensemble burn probability, VIIRS-derived support, and cell-level true-positive, false-positive, and false-negative regions.

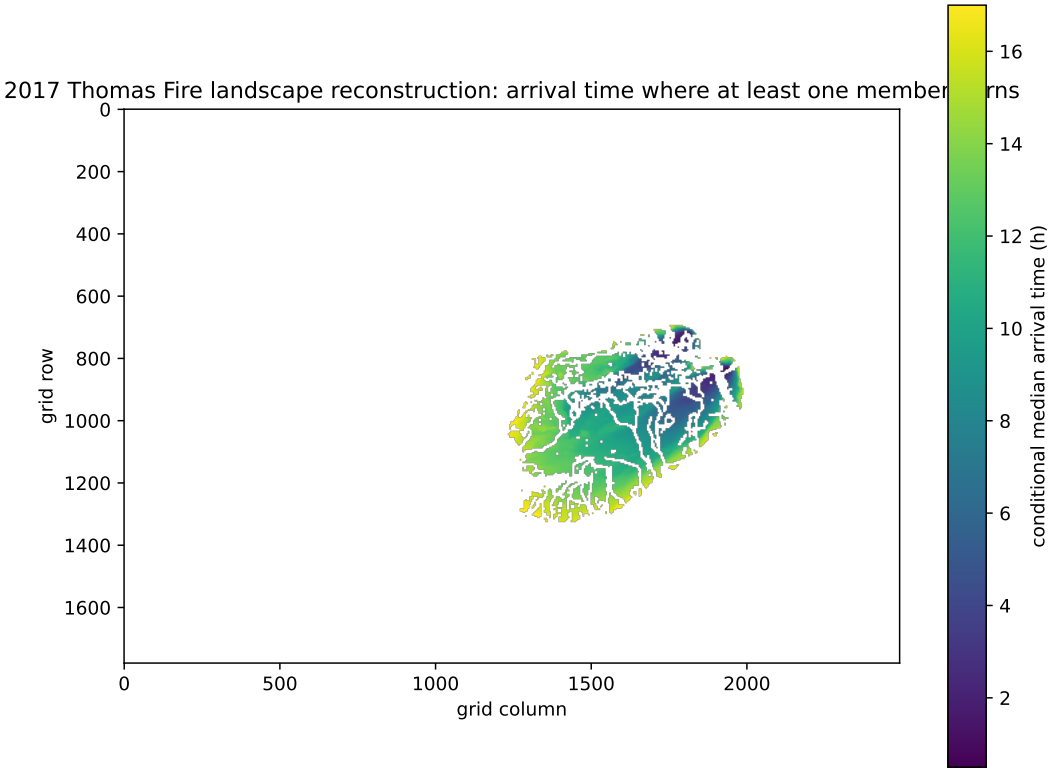


Figure 6: Conditional median arrival time among members that burn each cell by the comparison time. Conditioning is stated because it omits no-burn members.

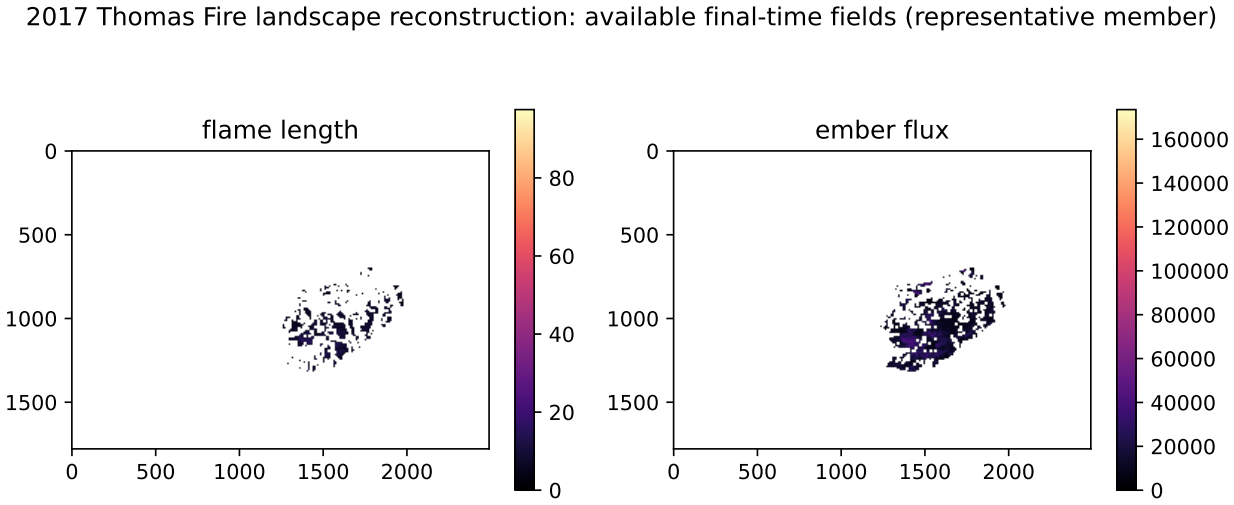


Figure 7: Applicable final-time two-dimensional fields found by the postprocessor. Omission of this panel means no recognized optional field was available; it does not imply a zero field.

The simulation requests flame length and ember flux in addition to arrival time. It does not request a structure-damage count or damage-time output, so a damaged-structures time series is not reported and must not be inferred from WUI input rasters alone.

8 Integrated statistics and calculated metrics

Metric	Current value
Jaccard overlap	0.6214
Precision	0.7985
Recall	0.7369
F_1 score	0.7665
Brier score	0.03336
Predicted/observed area ratio	0.9227
Predicted area (km^2)	266.8
Observation-support area (km^2)	289.1
Ensemble members found	50

Table 3: Calculated statistics. **MISSING** is intentionally conspicuous when the completeness requirement prevents evaluation.

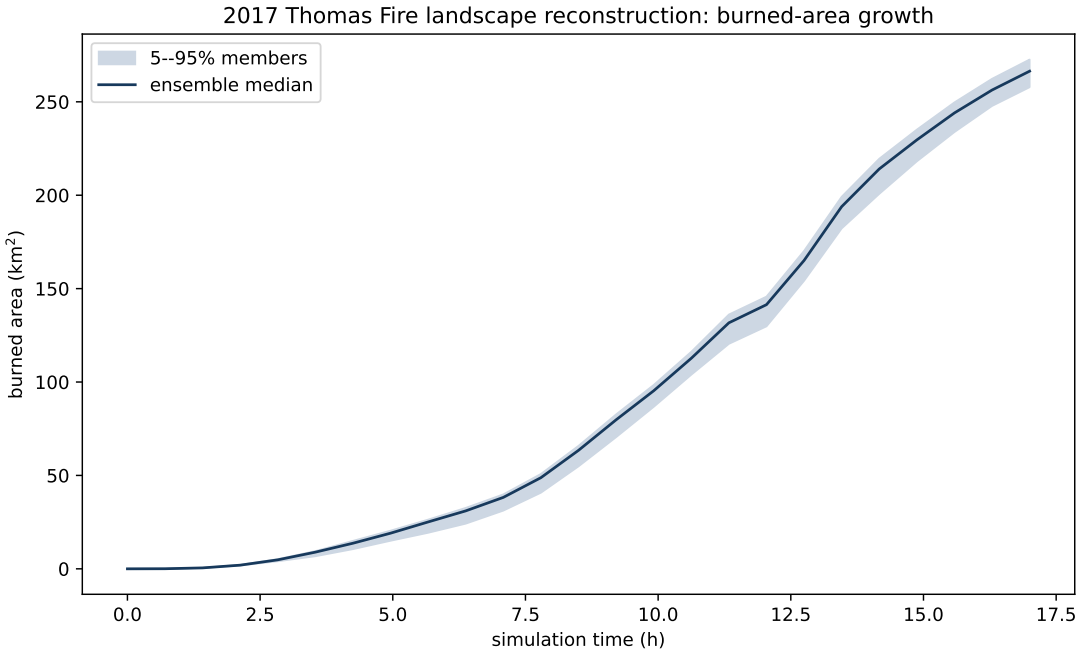


Figure 8: Median and 5th–95th percentile ensemble burned-area growth.

These values are compared with the observation-derived support and not with a stored ELMFIRE baseline. Because neither metric thresholds nor observation uncertainty bounds were declared independently, a populated table supports scientific characterization only.

9 Interpretation and uncertainty

A low Jaccard value with high recall and low precision indicates spatial overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory. An area ratio near one cannot establish

spatial agreement. The Brier score must be read with the footprint metrics because correct prediction of the large unburned domain may obscure errors near the fire.

The dominant limitations are incomplete input provenance, an observation product that samples thermal anomalies rather than perimeter, uncertain hull sensitivity, unknown event-specific calibration history, and absence of predeclared acceptance bounds. Model-form error, uncertain fuels and moisture, meteorological representation, ignition timing/location, spotting parameters, and WUI parameters are not separable using these aggregate metrics alone. The 50 archived perturbations characterize one configured ensemble; they do not by themselves quantify all epistemic uncertainty.

10 Conclusions and limitations

This case characterizes the agreement between a complete simulated ensemble and a spatial support derived from active-fire detections. Input statistics describe the forcing, spatial comparisons distinguish omission and commission, and integrated quantities provide context for the overlap metrics. Its current scientific result is **CHARACTERIZED**; the accompanying reason states whether the complete current ensemble supported quantitative characterization.

To support a defensible validation decision, future case owners must supply the upstream input citations and licenses, exact VIIRS collection/granule and acquisition provenance, the derivation or literature basis for ignition, spotting, WUI, and perturbation settings, disclosure of any calibration against this event, and acceptance thresholds fixed before inspecting the evaluated run. Until then, even a completed run is correctly labeled CHARACTERIZED, not PASS or FAIL.

2017 Tubbs Fire Landscape Reconstruction

1 Validation objective and scope

This case asks whether ELMFIRE can reconstruct the landscape-scale spatial growth of the 2017 Tubbs Fire Landscape Reconstruction through 12 hours when driven by the supplied landscape, weather, moisture, ignition, spotting, and wildland–urban interface (WUI) inputs. The comparison evidence is a time-filtered set of active-fire detections from the Visible Infrared Imaging Radiometer Suite (VIIRS). The case exercises the coupled surface- and crown-fire, level-set, firebrand, ember-ignition, Monte Carlo, and building-spread processes. It does not isolate or verify any single algorithm, and it does not establish predictive skill for events outside the three received reconstructions.

The scientific question is deliberately two-sided: does the ensemble overlap the observation-derived footprint, and does it do so without achieving overlap by excessive fire growth? Overlap, omission, commission, area, and probabilistic error are therefore reported together. The archive provides no independent, predeclared acceptance thresholds, so completed runs are *characterized*; they are not labeled PASS or FAIL.

2 Reference event, experiment, and observations

The configured event start, expressed in Pacific daylight time (PDT), is 8 October 2017 at 21:45 PDT. The comparison instant is 9 October 2017 at 09:45 PDT (16:45 UTC), with UTC indicating Coordinated Universal Time. The supplied observation dataset contains 1168 point records in the geographic coordinate system defined by EPSG:4326; 597 occur on or before the comparison cutoff. The observation attributes identify the instrument as VIIRS and are consistent with the National Aeronautics and Space Administration (NASA) Fire Information for Resource Management System (FIRMS) active-fire product. NASA describes VIIRS active-fire data as nominal 375-m thermal detections, not mapped fire perimeters. General product information is available from NASA FIRMS active-fire product documentation and NASA Earthdata VIIRS near-real-time product documentation.

The exact FIRMS collection, granules, download URL, acquisition date, filtering history, and license are not recorded in the case material. Consequently the NASA attribution is inferred from the observation attributes rather than established by complete provenance. Integrity checks identify the supplied observations, which remain unchanged during preprocessing.

For raster comparison, detections through the cutoff are projected to the simulation coordinate reference system (CRS) and enclosed by a concave hull with ratio 0.2 and no holes. If the installed geometry library lacks that operation, the postprocessor reports and uses a convex-hull fallback. This converts intermittent points into a comparison support; it is a declared comparison assumption, not an observed final perimeter. Cloud, smoke, overpass timing, geolocation error, false detections, and the hull choice remain observation uncertainties.

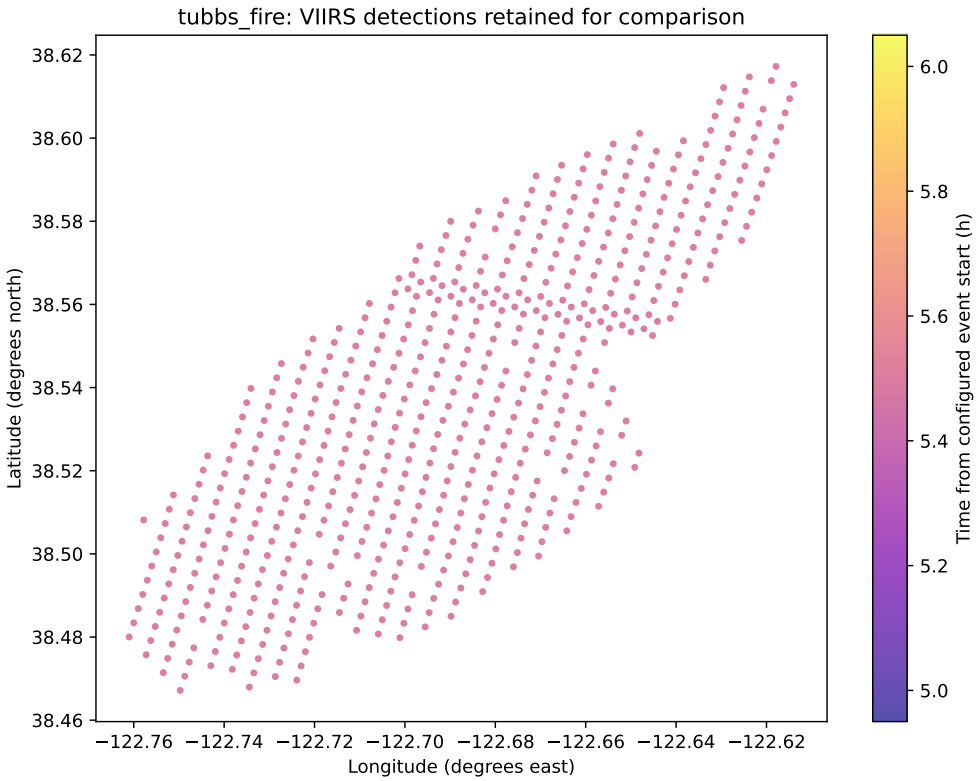


Figure 1: Supplied VIIRS detections retained through the comparison cutoff, colored by elapsed time from the configured event start.

3 Input data provenance and characterization

The supplied historical-fire reconstruction is identified by the SHA-256 checksum `e97bf430d251220 19211405f62f98979e929f80bc3f20d8c3b9f1f9f1a8a4955`. The extracted source data are retained unchanged, and their identities are checked before analysis. The collection's only supplied provenance note is "From Dwi." Upstream product names, providers, vintages, acquisition dates, processing history, and licenses for fuels, terrain, meteorology, moisture, WUI, and ignition data are unknown. The report therefore describes the archived values but does not infer unsupported product identities such as LANDFIRE or a specific weather analysis.

Field	Minimum	Mean	Standard deviation	Maximum
Elevation	15	164.8	130.8	722
Slope	0	8.329	8.383	47
Wind speed	0.0608	24.32	13.19	54.81
1-h moisture	4.225	5.335	0.3679	6.793
10-h moisture	5.225	6.335	0.3679	7.793
100-h moisture	6.225	7.335	0.3679	8.793

Table 1: Input statistics over valid cells and the 25 weather bands selected by the namelist. Wind speed is interpreted as 20-ft mph and dead-fuel moisture as percent from the archived flags and inspected current source; upstream metadata that would independently confirm these units is absent.

The 300-m weather rasters cover the same extent and CRS as the 30-m landscape but intentionally do not share its pixel grid. The preprocessor checks weather extent, band count, and CRS separately rather than treating the resolution difference as an error.

Aspect and wind direction are circular quantities. Their arithmetic means are not used as directional summaries; the circular means are 351.5° and 40.21°, respectively. Exact means and standard deviations use every valid selected value. Reported quantiles use a deterministic nearest-neighbor sample of at most 512×512 cells per band, and are labeled accordingly.

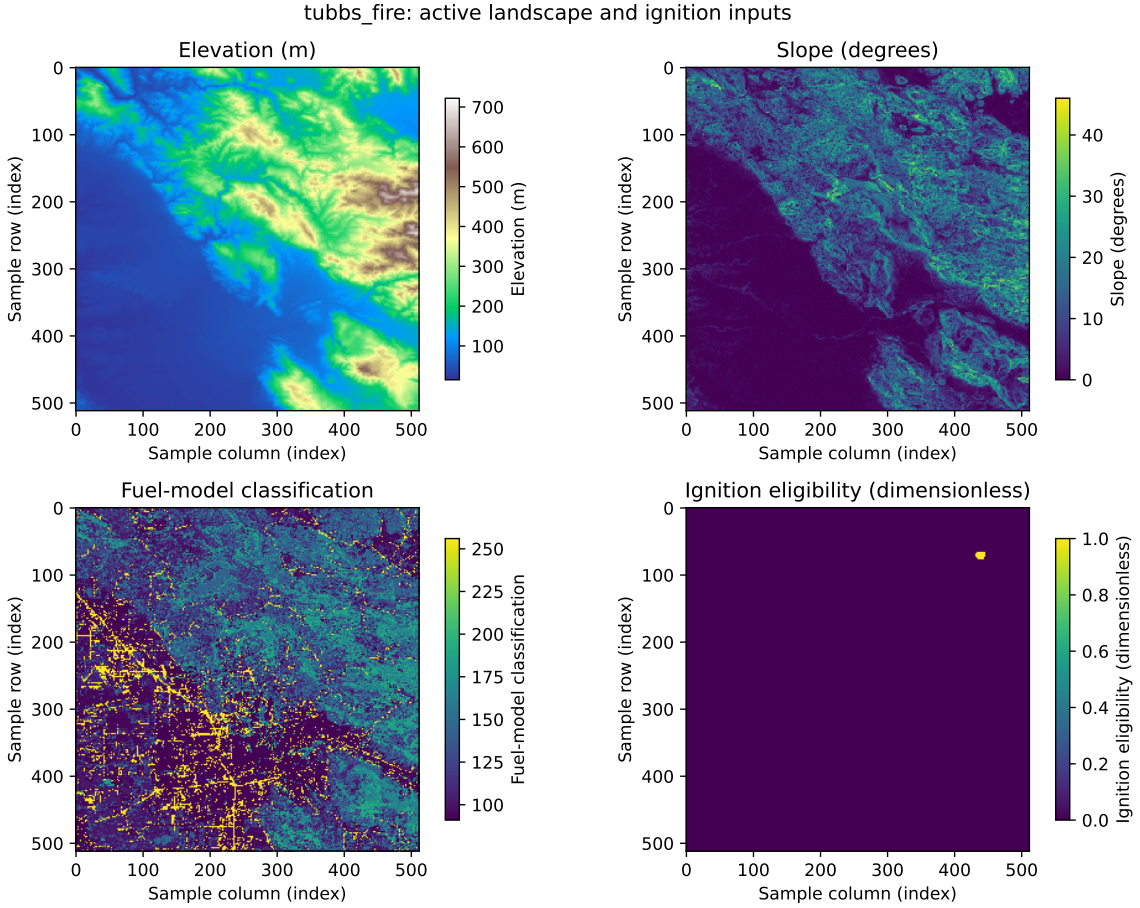


Figure 2: Active elevation, slope, fuel-model, and ignition-mask inputs. These plots characterize model forcing; they are not validation evidence.

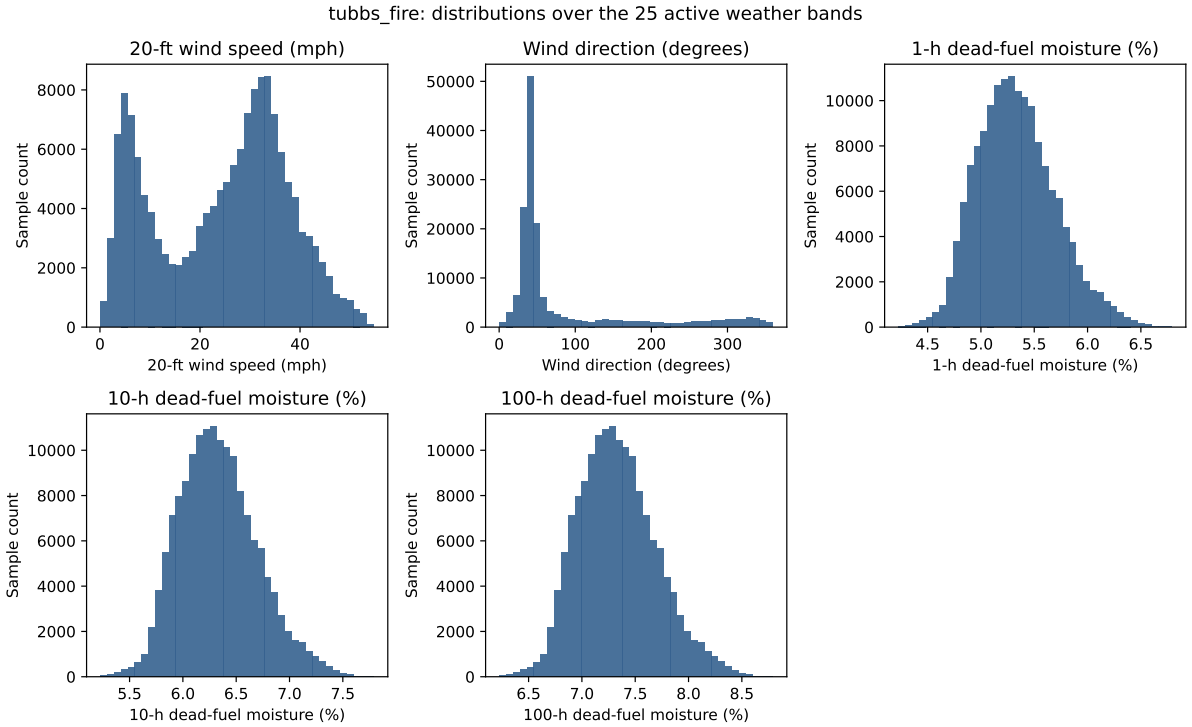


Figure 3: Distributions sampled deterministically from the 25 weather and dead-fuel-moisture bands selected by the namelist.

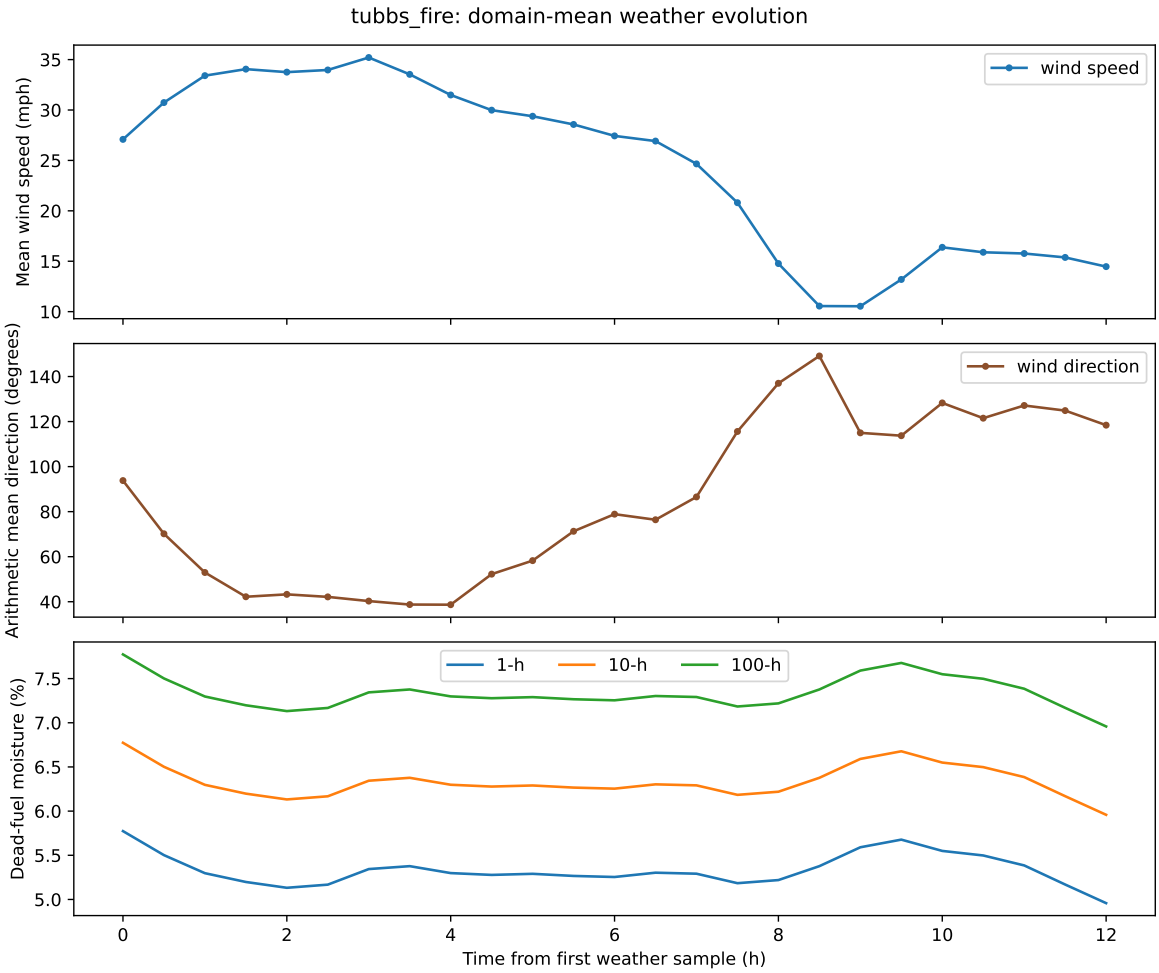


Figure 4: Exact domain-mean wind and dead-fuel-moisture values by active band.

4 Model configuration and rationale

The active domain is 700 by 900 cells at 30 m in the archived Lambert conformal conic CRS. Weather is 25-band, 300 m weather rasters at 30-minute intervals, and the ignition treatment is random ignition sampling from the archived ignition mask, with seed 2024. The simulation uses 50 members, a target Courant–Friedrichs–Lewy (CFL) number of 0.45, crown-fire model 2, and fire-power-proportional firebrand generation with empirical transport distance, Eulerian accumulation, and physical ignition. These physical values, seeds, uncertainty ranges, ignition choices, and WUI settings are retained from the received archive except for the documented removal of a contradictory fixed Tubbs ignition. In the available material they are *documented by configuration* but their scientific selection rationale is unsupported; they must not be described as independently derived or literature-justified.

The original settings were adapted only where required for compatibility with the inspected ELMFIRE formulation at source revision `efa0b8fa3935ea4cd30a2bd054c7e9b83db81163`, which already contained additional modifications when inspected. Obsolete domain-description and diagnostic settings were omitted, and the heat-release-rate ellipse adjustment was associated with the wildland–urban interface settings. The earlier spotting choices were expressed as fire-power-proportional generation, empirical transport distance, Eulerian accumulation, and physical ignition. Archived building spread type 3 was mapped to current University of California, Berkeley/University of Maryland building-

spread formulation (type 2). The contradictory archived fixed ignition was disabled because current ELMFIRE applies it in addition to the configured random ignition-mask sample, whereas this case specifies the latter alone. The required physical values are retained in the case specification; this compatibility review does not establish their scientific justification.

Choice	Evidence class	Basis and limitation
Landscape, weather, ignition, Monte Carlo, spotting, and WUI values	documented	Retained from the received archive; upstream derivation and event-specific tuning history are unknown.
Model-setting compatibility	derived	Required by the choices supported by the inspected model formulation; no physical values were tuned.
Comparison time	documented	Fixed in the case specification and consistent with the supplied reconstruction procedure; original scientific rationale is absent.
$p \geq 0.50$ footprint	selected for this comparison	Majority-ensemble interpretation used for reproducibility; it is not an acceptance threshold and sensitivity should be reported for scientific use.
VIIRS hull ratio 0.2	selected for this comparison	Compact cumulative support for point detections; not calibrated to an uncertainty-bounded perimeter.

Table 2: Provenance of consequential configuration and comparison choices.

5 Preprocessing and reproducibility

Preprocessing verifies the supplied reconstruction and observations by size and SHA-256 checksum. It checks that required landscape rasters share a grid, confirms that the namelist-selected weather rasters contain at least 25 bands, computes input and observation statistics, and regenerates the four input figures. The resulting statistics and figures characterize the source data without modifying the original evidence.

Simulation is followed by analysis of the complete ensemble. The declared default allocation is 50 Message Passing Interface (MPI) processes, one per ensemble member, without an additional coordinating process. The original simulation results remain available for examination. Reproducible input identity does not by itself guarantee result reproducibility: each run must also record the actual ELMFIRE revision, compiler, dependencies, and platform.

6 Validation metrics and comparison method

For member i, let $\tau_i(\mathbf{x})$ be the time-of-arrival raster and t_c the declared comparison time. The ensemble burn probability and thresholded footprint are

$$p(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[0 < \tau_i(\mathbf{x}) \leq t_c], \qquad P = \{\mathbf{x} : p(\mathbf{x}) \geq 0.50\}. \tag{1}$$

For the rasterized observation support O, intersection $I = |P \cap O|$, and valid-cell count M, the analysis reports

$$J = \frac{|P \cap O|}{|P \cup O|}, \qquad \text{precision} = \frac{I}{|P|}, \qquad \text{recall} = \frac{I}{|O|}, \tag{2}$$

$$F_1 = \frac{2\,\text{precision}\,\text{recall}}{\text{precision} + \text{recall}}, \qquad R_A = \frac{|P|}{|O|}, \qquad \text{Brier} = \frac{1}{M}\sum_{\mathbf{x}}(p - y)^2. \tag{3}$$

Jaccard is a symmetric footprint-overlap summary. Precision diagnoses commission/overgrowth, recall diagnoses omission/undergrowth, and F_1 balances those errors. The area ratio exposes size compensation that overlap alone can hide. The Brier score retains ensemble probabilities, although the large unburned domain can dominate it. Burned-area growth and conditional median arrival time add temporal context. VIIRS detections and their derived hull are the observational baseline; no previous simulation result is used as truth.

Metrics are calculated only when exactly 50 uniquely identified members reach the comparison time and share one grid and CRS. This completeness requirement prevents partial runs from being reported as scientific results. The static aspect raster defines the analysis-domain mask. Member-specific NoData cells in time-of-arrival outputs represent cells not burned by that member; they are therefore retained as zero burn outcomes rather than treated as a grid-mask mismatch.

7 Simulation results and field-level comparison

The current scientific status is **CHARACTERIZED**. Metrics are descriptive because no independent predeclared acceptance thresholds are available. Metrics and figures are generated only from the complete, current ensemble. If that ensemble is unavailable, the analysis reports the result as not evaluable rather than substituting legacy outputs or zero-valued metrics.

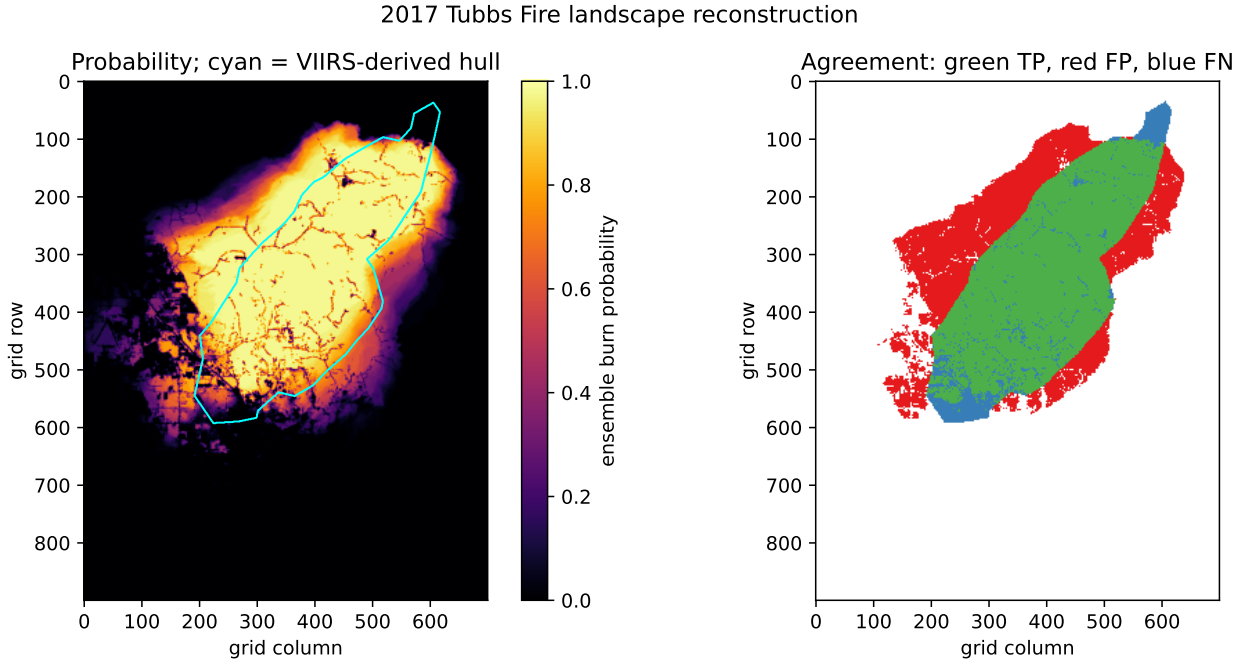


Figure 5: Ensemble burn probability, VIIRS-derived support, and cell-level true-positive, false-positive, and false-negative regions.

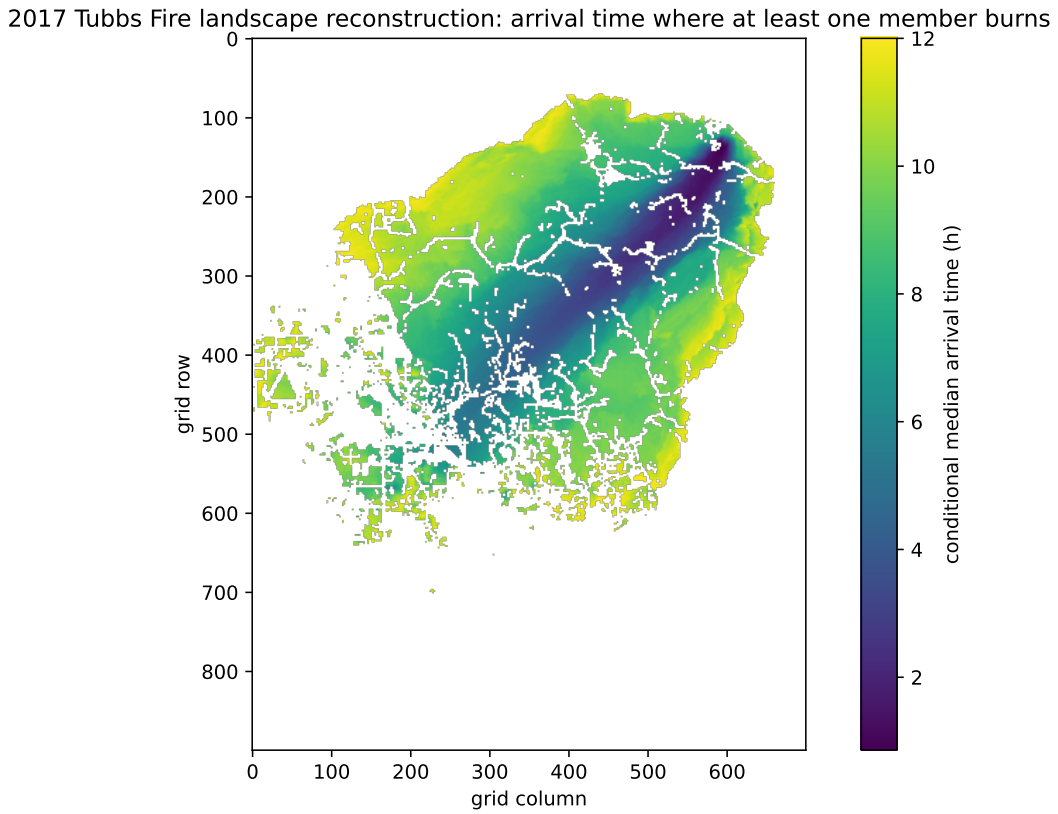


Figure 6: Conditional median arrival time among members that burn each cell by the comparison time. Conditioning is stated because it omits no-burn members.

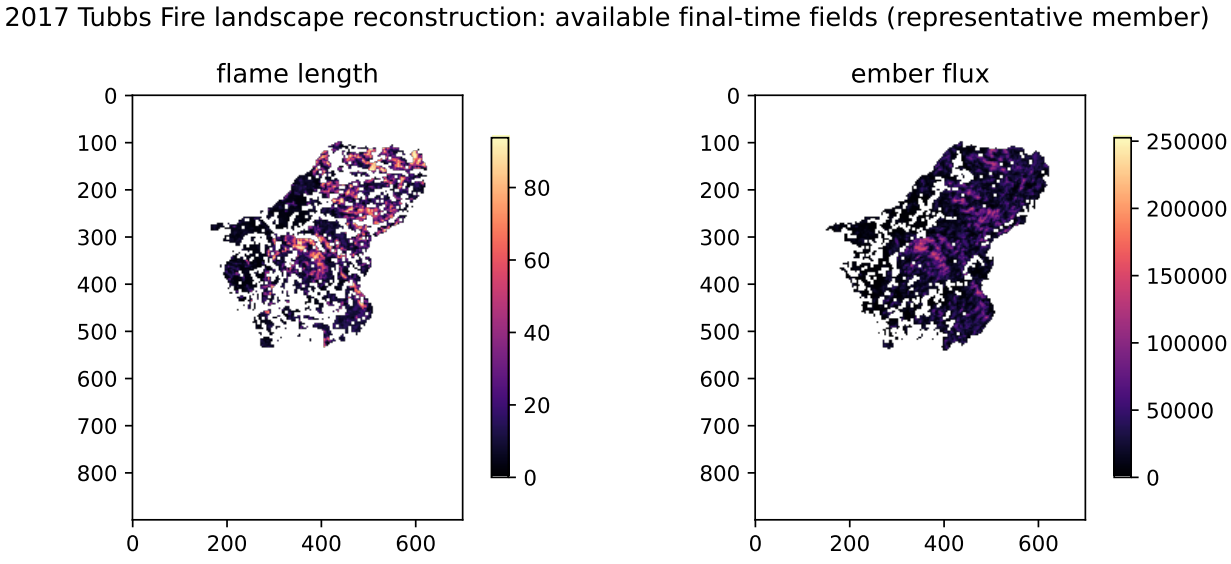


Figure 7: Applicable final-time two-dimensional fields found by the postprocessor. Omission of this panel means no recognized optional field was available; it does not imply a zero field.

The simulation requests flame length and ember flux in addition to arrival time. It does not request a structure-damage count or damage-time output, so a damaged-structures time series is not reported

and must not be inferred from WUI input rasters alone.

8 Integrated statistics and calculated metrics

Metric	Current value
Jaccard overlap	0.555
Precision	0.6051
Recall	0.8702
F_1 score	0.7138
Brier score	0.09008
Predicted/observed area ratio	1.438
Predicted area (km^2)	133.8
Observation-support area (km^2)	93.02
Ensemble members found	50

Table 3: Calculated statistics. **MISSING** is intentionally conspicuous when the completeness requirement prevents evaluation.

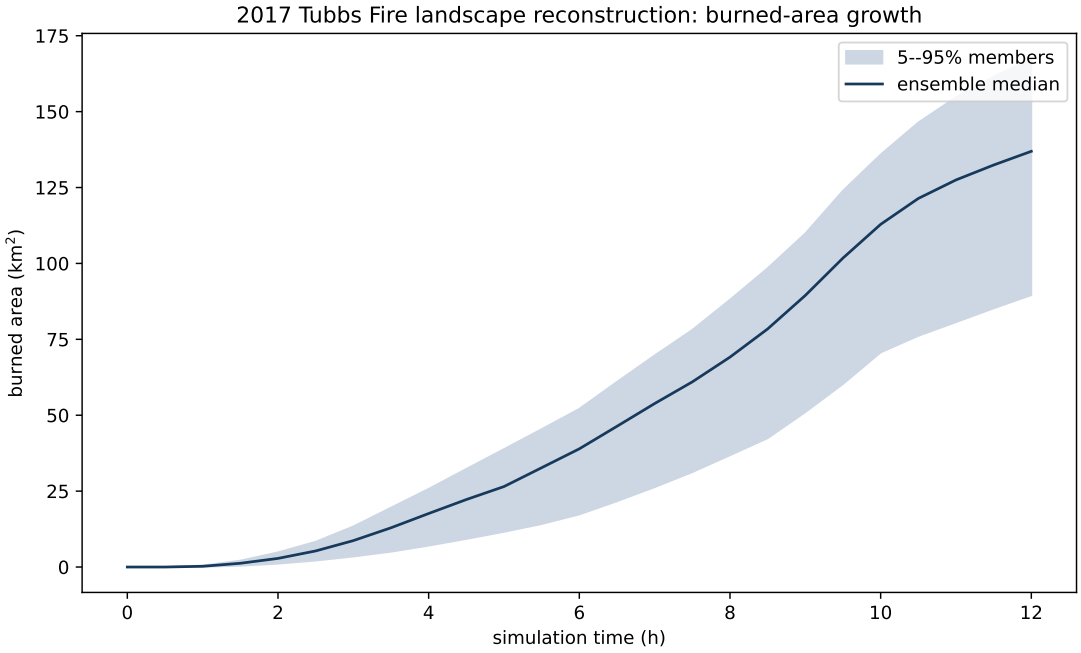


Figure 8: Median and 5th–95th percentile ensemble burned-area growth.

These values are compared with the observation-derived support and not with a stored ELMFIRE baseline. Because neither metric thresholds nor observation uncertainty bounds were declared independently, a populated table supports scientific characterization only.

9 Interpretation and uncertainty

A low Jaccard value with high recall and low precision indicates spatial overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory. An area ratio near one cannot establish spatial agreement. The Brier score must be read with the footprint metrics because correct prediction of the large unburned domain may obscure errors near the fire.

The dominant limitations are incomplete input provenance, an observation product that samples thermal anomalies rather than perimeter, uncertain hull sensitivity, unknown event-specific calibration history, and absence of predeclared acceptance bounds. Model-form error, uncertain fuels and moisture, meteorological representation, ignition timing/location, spotting parameters, and WUI parameters are not separable using these aggregate metrics alone. The 50 archived perturbations characterize one configured ensemble; they do not by themselves quantify all epistemic uncertainty.

10 Conclusions and limitations

This case characterizes the agreement between a complete simulated ensemble and a spatial support derived from active-fire detections. Input statistics describe the forcing, spatial comparisons distinguish omission and commission, and integrated quantities provide context for the overlap metrics. Its current scientific result is **CHARACTERIZED**; the accompanying reason states whether the complete current ensemble supported quantitative characterization.

To support a defensible validation decision, future case owners must supply the upstream input citations and licenses, exact VIIRS collection/granule and acquisition provenance, the derivation or literature basis for ignition, spotting, WUI, and perturbation settings, disclosure of any calibration against this event, and acceptance thresholds fixed before inspecting the evaluated run. Until then, even a completed run is correctly labeled CHARACTERIZED, not PASS or FAIL.